# Types of Screen Time and Developmental Outcomes Among Children Aged 0–6 Years in Thailand: A Cross■Sectional Study

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## Resumo (Abstract)

ABSTRACT Introduction Excessive screen time during early childhood can affect developmental outcomes. However, evidence on the differential effects of screen time types in Thailand remains limited. In this study, we examined associations between different types of screen time and developmental outcomes among children aged 0–6 years in Thailand. Methods This cross■sectional study was conducted in community kindergartens and primary healthcare centres in Thailand. A multistage area■selection procedure was used to select the study settings, followed by convenience sampling of caregiver–child pairs. Data on screen time and related characteristics were collected through structured caregiver interviews. Children's developmental outcomes were assessed using the Developmental Surveillance and Promotion Manual by trained public health practitioners. Adjusted multivariable logistic regression models were used to examine associations between different types of screen time and suspected developmental delays. Results In total, 465 caregiver–child pairs were included. The mean age of children was 41.97 (SD 19.76) months. The median daily screen time was 2.25 h (IQR 1.83–3.00), with 78.9% exceeding recommended limits. Each additional hour of total screen time was associated with increased odds of suspected developmental delays across developmental domains, particularly receptive language delay (aOR 1.52, 95% CI 1.30–1.79) and gross motor delay (aOR 1.49, 95% CI 1.21–1.84). Recreational screen time showed the strongest associations, particularly for receptive language delay (aOR 2.39, 95% CI 1.73–3.30) and personal–social delay (aOR 2.27, 95% CI 1.68–3.08). In contrast, educational screen time was associated with lower odds of receptive language delay (aOR 0.45, 95% CI 0.23–0.88) and expressive language delay (aOR 0.40, 95% CI 0.16–0.99), whereas communication■based screen time was associated with lower odds of personal–social delay (aOR 0.14, 95% CI 0.02–0.95). Conclusions Higher total and recreational screen time were associated with increased odds of suspected developmental delays, whereas educational and communication■based screen time were associated with lower odds in specific domains. Limiting recreational screen time and promoting age■appropriate, high■quality, interactive screen use may support child development.

## Introduction

Optimal movement behaviours, including limited sedentary screen time, regular active play, and adequate, high■quality sleep, are fundamental to healthy growth and development in children aged under five years. The World Health Organisation (WHO) recommends no screen exposure for children younger than 2 years and limiting sedentary screen time to no more than 1 h/day for children aged 2–5 years (World Health Organisation 2019 ), and limiting recreational screen time to no more than 2 h/day for older children (Australian Institute of Family Studies 2026 ). Despite these recommendations, screen use among young children frequently exceeds guideline limits worldwide, with only 24.7% of children younger than 2 years and 35.6% of those aged 2–5 years meeting these recommendations (McArthur et al. 2022 ). Increasing access to digital media and family screen■use behaviours may further contribute to high levels of screen exposure among young children

(Angkittisawat et al. 2022 ; Tansriratanawong et al. 2017 ). These trends are concerning because early childhood is a critical period for brain development and lifelong well■being. Growing evidence indicates that excessive screen time is associated with developmental delays across language, socio■emotional, and motor domains (Ahmer et al. 2025 ; Madigan et al. 2020 ; Webster et al. 2019 ).

Previous studies conducted in multiple countries have consistently reported associations between screen exposure and developmental outcomes in early childhood. Evidence from high■income settings, including the United States, Canada, and Japan, indicates that excessive screen time is linked to greater vulnerability across language, cognitive, communication, motor, and socio■emotional domains (Kerai et al. 2022 ; Qu et al. 2023 ; Takahashi et al. 2023 ). These associations may be particularly pronounced during early childhood because of the heightened neurodevelopmental sensitivity of this developmental period (Qu et al. 2023 ). Neuroimaging studies further suggest that greater screen exposure is related to reduced white matter integrity in brain regions involved in language and executive functioning (Hutton et al. 2020 ). Importantly, developmental outcomes appear to vary according to the type and context of screen use. Passive screen use (e.g., television viewing) is generally associated with poorer outcomes, whereas high■quality educational and interactive content, combined with caregiver co■engagement, may support language and executive functioning (Lakicevic et al. 2025 ; Madigan et al. 2020 ). Collectively, these findings underscore the importance of considering both the quantity and quality of screen use in early childhood.

Screen use among young children in Thailand is widespread, with 30.9% exceeding 1 h/day and 55.2% initiating screen exposure before 2 years of age (Angkittisawat et al. 2022 ). These patterns are concerning because both excessive screen time and early screen initiation have been associated with an increased risk of developmental delays (Takahashi et al. 2023 ). A recent Thai study reported that longer screen time was associated with delays in fine motor, cognitive, and expressive language development, whereas screen content was not significantly associated with developmental outcomes (Panatthanang and Leesri 2026 ). This finding contrasts with evidence suggesting that developmental outcomes may vary according to the type and context of screen use, with educational and communication■based media being associated with better language and socio■emotional outcomes (Glick et al. 2022 ). However, few studies have systematically examined different types of screen use across multiple developmental domains. Evidence remains particularly limited in Thailand, while most studies from other countries have focused primarily on total screen time (Kerai et al. 2022 ; Qu et al. 2023 ; Takahashi et al. 2023 ). By examining recreational, educational, and communication■based screen time separately, this study extends existing evidence beyond total screen time and advances the current understanding of how different types of screen time relate to specific domains of early childhood development, thereby providing further evidence to inform future screen■use guidance and longitudinal research. Given the rapid brain development that occurs during early childhood (Nelson and Gabard■Durnam 2020 ) and the recommendations of Thailand's Developmental Surveillance and Promotion Manual (DSPM), which assesses developmental milestones from birth to 6 years of age (Ministry of Public Health 2023 ), this study examined associations between different types of screen time (recreational, educational, and communication■based) and developmental outcomes among children aged 0–6 years in Thailand. We hypothesised that recreational screen time would be associated with higher odds of developmental delays, whereas educational and communication■based screen time would be associated with lower odds of developmental delays among children aged 0–6 years.

# Methods

## Study Design

This cross■sectional study included caregiver–child pairs with children aged 0–6 years in Thailand from 1 February to 31 July 2025. The study was reported in accordance with the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) guidelines (see Supplementary File S1 ).

## Participants and Setting

A multistage sampling approach was used to enhance geographic representation across Thailand. In the first stage, five provinces were randomly selected from the four regions of Thailand and the Bangkok Metropolitan Region. In the second stage, three districts were randomly selected from each province ( n = 15). In the third stage, one subdistrict was selected from each district ( n = 15). Within the selected areas, caregiver–child pairs were recruited from community kindergartens and primary healthcare centres using convenience sampling. Eligible caregiver–child pairs who attended routine child health services or early childhood programmes were invited to participate. A flow diagram illustrating the participant selection process is provided in Supplementary File S2 .

Participants were caregiver–child pairs. The inclusion criteria were: (1) children aged 0–6 years; (2) primary caregivers responsible for the child's daily care since birth for children aged < 6 months, or for at least the preceding 6 months for children aged ≥ 6 months; (3) caregivers aged ≥ 20 years; and (4) caregivers who were able to communicate in Thai and provide informed consent. Exclusion criteria were: (1) children with a clinically diagnosed developmental delay or neurodevelopmental disorder before study recruitment, as documented in medical records or reported by caregivers; (2) children with serious medical conditions affecting developmental assessments; and (3) caregivers with mental health conditions or cognitive impairment affecting response reliability.

## Ethical Considerations and Data Collection

Ethical approval was obtained from the Institutional Review Board of Taipei Medical University (No. N202409048) and the Research Ethics Committee of the Faculty of Nursing, Chiang Mai University (No. 2567■FULL025). Written informed consent was obtained from caregivers on behalf of themselves and their children before participation. Data were collected through face■to■face interviewer■administered structured questionnaires and developmental assessments conducted in private settings by public health practitioners. Before data collection, all assessors completed standardised training on study procedures, questionnaire administration, and Developmental Surveillance and Promotion Manual (DSPM) administration and were certified according to the Ministry of Public Health protocols. Standardised DSPM assessment and scoring procedures were implemented across all study sites to ensure assessment quality and consistency, minimise inter■assessor variation, and reduce interviewer bias. Participants were assured that their responses would remain confidential and were informed of their right to withdraw from the study at any time without consequences.

## Sample Size

The sample size was estimated using Cochran's method for estimating a proportion (Cochran 1977 ). Assuming a 95% confidence level ( Z = 1.96), a prevalence of suspected developmental delay of 0.26 (Thaineua et al. 2024 ), and a precision ( d ) of 0.05, the required sample size was 296

caregiver–child pairs. Given the multistage sampling design, a design effect (DEFF) of 1.3 was applied based on an assumed intraclass correlation coefficient (ICC) of 0.01 and an average cluster size of 30 participants (Stuart et al. 2020 ), increasing the required sample size to 385 caregiver–child pairs. After accounting for a 10% non■response rate, the final required sample size was 424 caregiver–child pairs.

## Measurements

## Sociodemographic Characteristics

A questionnaire was used to collect sociodemographic information on child and caregiver characteristics. Child characteristics included age, sex, and birth history. Caregiver characteristics included age, relationship to the child, educational attainment, marital status, and household income.

## Screen Use Characteristics

A structured questionnaire was developed based on established guidelines for children's screen use (American Academy of Paediatrics 2016 ; World Health Organisation 2019 ). Primary caregivers reported their child's age at first screen exposure, number and types of devices, co■viewing patterns, and screen use on smartphones, tablets, televisions, computers, and gaming consoles during the preceding 7 days. Screen use was assessed separately for typical weekdays and weekend days and categorised as recreational (e.g., television viewing, online videos, and digital games), educational (e.g., educational applications and learning videos), or communication■based screen time (e.g., video calls). Caregivers reported the average duration of each category in hours and minutes per day based on a typical weekday and weekend day during the preceding 7 days. Daily screen time (h/day) was calculated as [(weekday screen time × 5) + (weekend screen time × 2)] /7 (McArthur et al. 2022 ). Excessive screen time was defined according to age■specific recommendations from the World Health Organisation ( 2019 ) and the Australian Institute of Family Studies ( 2026 ): no screen time for children aged 0–23 months, ≤ 1 h/day for those aged 24–59 months, and ≤ 2 h/day for those aged 60–72 months. Content validity was evaluated by six experts (item■level content validity index [I■CVI], 0.83–1.00; scale■level content validity index, average method [S■CVI/Ave], 0.98), indicating good content validity.

## Developmental Surveillance and Promotion Manual (DSPM)

Child development was assessed using the DSPM, a nationally standardised developmental screening tool developed by the Ministry of Public Health, Thailand, for children aged 0–6 years. The DSPM evaluates five developmental domains: gross motor, fine motor, receptive language, expressive language, and personal–social development using age■specific dichotomous items (pass/fail). For each domain, children who failed one or more age■appropriate items were classified as having suspected developmental delay, generating separate binary outcomes for logistic regression analyses. Assessments incorporated both caregiver reports and direct observation of the child's performance. Previous studies have reported good psychometric properties (item■objective congruence [IOC] > 0.50; inter■observer reliability > 0.80; sensitivity 96.0%; specificity 64.6%) (Sirithongthaworn 2018 ). Inter■rater reliability in the present study was high (Cohen's $\kappa = 0.87$).

## Data Analysis

Statistical analyses were performed using Stata version 18 (StataCorp LLC, College Station, TX, USA). Normality was assessed using the Kolmogorov–Smirnov test. Continuous variables were

presented as mean (standard deviation [SD]) or median (interquartile range [IQR]), as appropriate, and categorical variables were presented as frequencies and percentages. Mean screen time and 95% confidence intervals (CIs) were plotted across age groups. Associations between participant characteristics and suspected developmental delays were examined using chi■squared ( $\chi$ 2 ) or Fisher's exact tests, as appropriate. Associations between different types of screen time and suspected developmental delays in each developmental domain were analysed using multivariable logistic regression, with adjusted odds ratios (aORs) and 95% CIs reported. Models were adjusted for child age, sex, low birth weight, preterm birth, caregiver education, marital status, household income, age at first screen exposure, passive device use, and co■viewing. Multicollinearity was assessed using variance inflation factors (VIFs), with all values below 5 indicating no substantial multicollinearity. Statistical significance was set at $p < 0.05$. Participants with more than 20% missing data were excluded. Because missing data were low (< 5%) and Little's Missing Completely at Random (MCAR) test was non■significant ( $p > 0.05$), complete■case analysis was applied.

# Results

## Participant Characteristics

Of the 500 caregiver–child pairs approached, 35 were excluded: 18 withdrew consent, 10 had more than 20% missing data, and 7 met the exclusion criteria. A total of 465 caregiver–child pairs were included in the analysis. The mean age of the children was 41.97 (SD 19.76) months, and 56.3% were female. Low birth weight and preterm birth were reported in 6.7% and 5.6% of children, respectively. Caregivers had a mean age of 38.30 (SD 12.50) years. Most caregivers were mothers (65.0%), followed by grandparents (21.7%). The majority were married (91.8%) and had an upper secondary education or above (63.9%). Approximately three■quarters of households (76.4%) had low or middle household incomes (Table 1 ).

## Screen Use Characteristics and Developmental Outcomes

Children began using digital devices at a mean age of 21.37 (SD 11.29) months. Median total screen time was 2.25 h/day (IQR 1.83–3.00), and 78.9% exceeded age■specific screen time recommendations. Nearly half of the children used passive devices (47.3%), and 51.2% had no caregiver co■viewing. The prevalence of suspected developmental delay was 12.5%, with fine motor (5.8%) and receptive language (5.6%) showing the highest prevalence (Table 2 ). Mean total screen time increased with age, from approximately 1.0 h/day among children aged 0–12 months to 3.4 h/day among those aged 61–72 months. Recreational screen time was consistently the highest across age groups, followed by educational screen time, whereas communication■based screen time remained the lowest (Figure 1 ).

## Associations Between Participant Characteristics and Suspected Developmental Delays

Bivariate analyses showed several significant associations with suspected developmental delays across the developmental domains (Table 3 ). Child age group was associated with gross motor and receptive language delays, whereas male sex and preterm birth were associated with fine motor delay. Low birth weight was associated with receptive and expressive language delays (all $p < 0.05$). Among caregiver characteristics, educational attainment, marital status, and household income were associated with expressive language delay, receptive language and personal–social delays, and fine motor delay, respectively (all $p < 0.05$). Regarding screen■use characteristics, screen exposure

before 2 years of age, passive device use, and no caregiver co■viewing were associated with language■related delays, whereas passive device use was additionally associated with personal–social delay (all $p < 0.05$). Caregiver age and interactive device use were not significantly associated with any developmental domain.

## Associations Between Screen Time and Suspected Developmental Delays

Multivariable logistic regression analyses showed that each additional hour of total screen time was associated with higher odds of suspected developmental delays across all developmental domains (Table 4 ). Among all screen■use categories, recreational screen time demonstrated the strongest and most consistent associations across all developmental domains, with the largest effects observed for receptive language (aOR 2.39, 95% CI 1.73–3.30) and personal–social delays (aOR 2.27, 95% CI 1.68–3.08). In contrast, educational screen time was associated with lower odds of receptive (aOR 0.45, 95% CI 0.23–0.88) and expressive language delays (aOR 0.40, 95% CI 0.16–0.99), whereas communication■based screen time was associated with lower odds of personal–social delay (aOR 0.14, 95% CI 0.02–0.95). No other significant associations were observed.

## Discussion

The present findings indicate that the developmental impact of screen use among Thai children aged 0–6 years varied according to the type of screen use and was not explained solely by screen duration. Developmental vulnerability was also associated with biological, sociodemographic, and family■related factors, highlighting the importance of considering multiple influences when evaluating the developmental effects of screen use in early childhood.

## Screen Time Patterns in Early Childhood

Excessive screen time was highly prevalent among Thai children aged 0–6 years, with only 21.1% meeting recommended limits, a proportion lower than global estimates reported for children younger than 5 years (McArthur et al. 2022 ). This difference may reflect the broader age range included in the present study, as screen use increases with age and early initiation may reinforce habitual screen■use patterns (Choi et al. 2023 ; Dhar et al. 2026 ). Of particular concern, more than two■fifths of infants exceeded screen time limits despite recommendations discouraging screen use before 2 years of age (World Health Organisation 2019 ). Our findings from communities across multiple regions of Thailand are consistent with those of previous Thai studies, showing that screen exposure commonly begins within the first two years of life, screen time increases throughout early childhood, and children are exposed to multiple forms of media (Angkittisawat et al. 2022 ; Supanitayanon et al. 2020 ; Tansriratanawong et al. 2017 ). Similar patterns of screen use have also been reported in low■ and middle■income countries, where screen■related movement behaviours were comparable among young children living in urban and rural communities (Nusurupia et al. 2024 ). Additionally, approximately half of the children used passive devices or had no caregiver co■viewing, potentially limiting opportunities for interactive engagement and active learning (Mallawaarachchi et al. 2024 ; Tansriratanawong et al. 2017 ). This pattern may partly reflect Thailand's multigenerational family structure, in which grandparents frequently assume caregiving responsibilities but may face challenges in providing alternatives to screen■based activities (Boonlue et al. 2026 ; Budden et al. 2024 ). These challenges, together with inconsistent adherence to recommended screen■use practices among caregivers, may contribute to excessive or unsupervised screen use among young children (Angkittisawat et al. 2022 ).

## Sociodemographic and Biological Covariates Influencing Development

Beyond screen time, several biological and sociodemographic factors were associated with developmental vulnerability. Preterm birth and low birth weight were linked to fine motor and language delays, consistent with evidence on the lasting neurodevelopmental effects of adverse perinatal conditions (Chung et al. 2020 ). Male sex was associated with fine motor delay, consistent with previous reports of sex differences in early motor development (Strooband et al. 2021 ). Children aged 25–48 months showed higher rates of gross motor and receptive language delays, consistent with the heightened developmental sensitivity of this age period (Nelson and Gabard■Durnam 2020 ). Lower household income, lower caregiver education, and caregiver marital disruption were associated with fine motor, language, and personal–social delays, underscoring the importance of family resources and environmental stability for early development (Kang et al. 2025 ; Strooband et al. 2021 ). Additionally, early screen exposure, passive device use, and the absence of caregiver co■viewing were associated with language■related and personal–social delays. These factors may reduce opportunities for verbal interaction and social engagement during critical developmental periods (Lakicevic et al. 2025 ; Madigan et al. 2020 ), consistent with evidence highlighting the role of caregiver screen■management practices in shaping children's media use (Boonmun et al. 2023 ).

## Overall Associations Between Screen Time and Developmental Outcomes

Higher total daily screen time was associated with 17%–52% greater odds of suspected developmental delays across motor, language, and personal–social domains in early childhood, consistent with previous studies from Thailand and other countries reporting a dose–response relationship between screen time and developmental risk, despite differences in sociocultural and caregiving contexts (Kerai et al. 2022 ; Panatthanang and Leesri 2026 ; Qu et al. 2023 ; Takahashi et al. 2023 ). Recreational screen time was the predominant form of screen use and showed the strongest associations with developmental delays. These associations may reflect its largely passive and entertainment■oriented nature, which can displace physical play, learning opportunities, and caregiver–child interactions (Pons et al. 2022 ; Veraksa et al. 2021 ). Additionally, highly stimulating content may contribute to attentional difficulties and reward■driven behaviours that affect multiple developmental domains (Chung et al. 2020 ; Lissak 2018 ). In contrast, educational screen time was associated with lower odds of language■related delays, consistent with evidence that high■quality educational or interactive content and caregiver–child co■viewing support vocabulary acquisition, language comprehension, and verbal expression (Lakicevic et al. 2025 ; Madigan et al. 2020 ). Furthermore, communication■based screen time was associated with lower odds of personal–social delays, potentially because video calls provide contingent verbal and non■verbal interactions that resemble natural social exchanges (Bodur et al. 2023 ). Collectively, these findings suggest that the developmental impact of screen use in early childhood depends on both the duration and type of media exposure.

## Strengths and Limitations

This study has several strengths. It differentiated between types of screen time and examined domain■specific developmental outcomes, providing evidence beyond total screen time. The multisite design and standardised assessments by trained public health practitioners strengthened the robustness of the findings. However, several limitations should be considered. First, the

cross■sectional design precludes establishing temporal relationships between screen time and developmental outcomes; therefore, causality and reverse causation cannot be ruled out. Second, screen time was caregiver■reported and may have been subject to recall bias, although standardised data collection procedures likely minimised this risk. Third, the study was conducted in community■based settings in Thailand, which may limit the generalisability to populations with different sociocultural contexts.

## Implications for Practice and Research

These findings suggest that screen■use guidance should move beyond screen time duration alone to emphasise the purpose, content, and social context of media use. In practice, efforts should focus on reducing recreational, passive, and solitary screen use, particularly when initiated at an early age, while promoting age■appropriate educational and communication■based media use with caregiver engagement. Future longitudinal studies are needed to clarify temporal relationships and examine how media characteristics, interactivity, and caregiver involvement influence developmental outcomes using objective measures and including more diverse populations.

## Conclusions

Total and recreational screen time were linked to increased odds of suspected developmental delays across motor, language, and personal–social domains in Thai children aged 0–6 years, whereas educational and communication■based screen time were linked to lower odds of language■ and personal–social■related delays. These findings suggest that developmental outcomes depend not only on the amount of screen use but also on the type and context of media use. Family■centred interventions should prioritise reducing recreational screen time while promoting age■appropriate, high■quality educational and communication■based media use with active caregiver engagement to support healthy early childhood development.

## Author Contributions

Sukritta Jaichomcheun: Conceptualization, Methodology, Formal analysis, Data curation, Project administration, Writing – original draft. Wachira Suriyawong: Project administration, Validation, Writing – review and editing. Pataporn Bawornthip: Investigation, Validation, Writing – review and editing. Su■Ru Chen: Conceptualization, Methodology, Supervision, Writing – review and editing.

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## Conflicts of Interest

The authors declare no conflicts of interest.

## Supporting information